Retail Customer Retention Analytics – Target

Project Overview

This project focuses on building a Customer Retention Analytics Dashboard for Target Corporation using Power BI. The objective is to analyze customer churn, retention behavior, loyalty effectiveness, promotion impact, store/channel performance, and customer lifetime value (CLV).

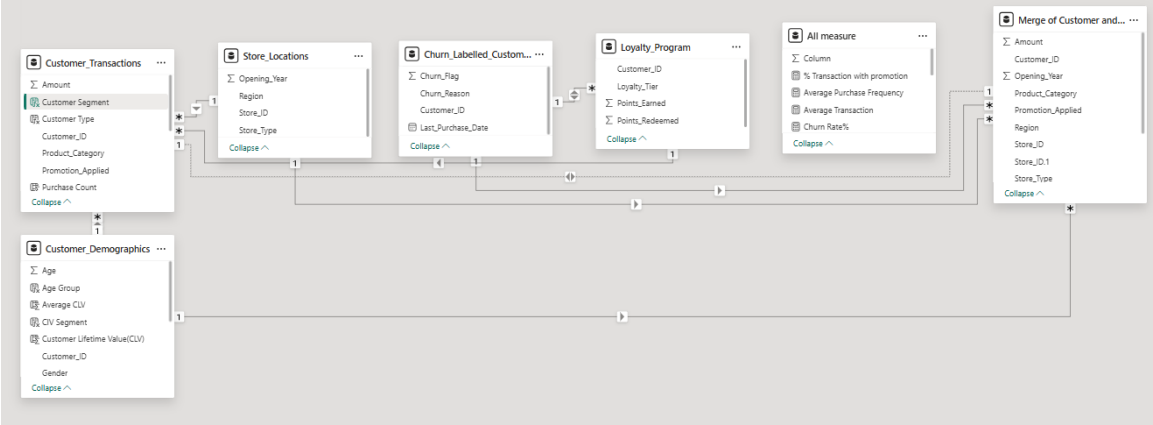
Task 1: Data Modeling & Cleaning

Datasets were loaded into Power BI using Power Query. Data cleaning involved removing duplicates, handling missing values, and correcting data types. Calculated columns were created for membership duration and transaction date hierarchy. Relationships were established across all tables.

Screenshot 1.2: Calculated Columns (Membership Duration, Year, Month)

Membership_Duration
854
2285
1019
1633
1194
2607
2271
2854
2184
2384
2436
2882
784
2545
2384
1574
2027
1879
1883
1263
1001

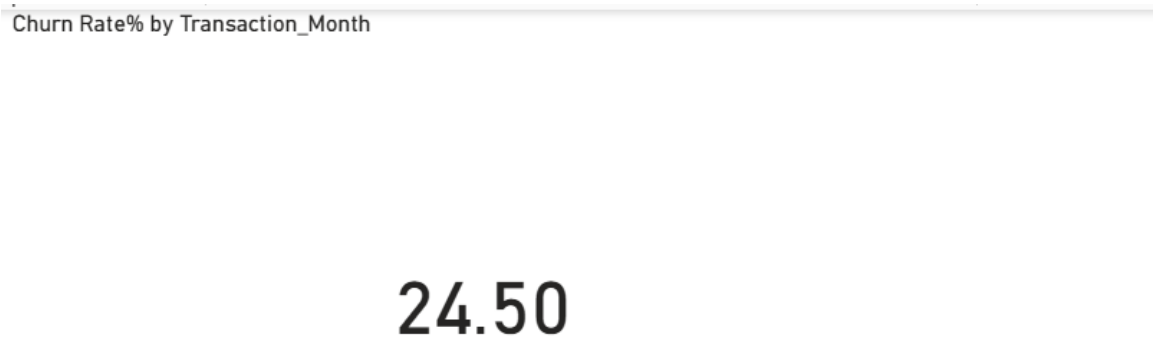
Screenshot 1.3: Data Model View showing table relationships



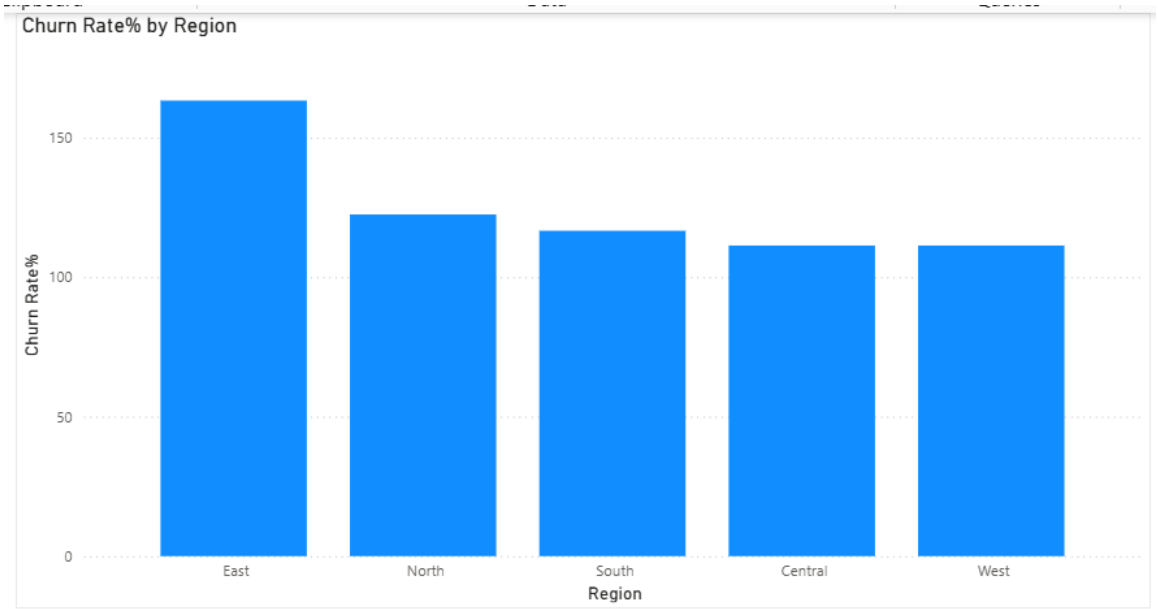
Task 2: Churn & Retention Metrics

A churn rate KPI was calculated using DAX. Churn analysis was conducted across Region, Income Level, Preferred Channel, and Loyalty Tier. A funnel chart was used to represent customer movement from total customers to repeat and churned customers.

Screenshot 2.1: Churn Rate KPI Card



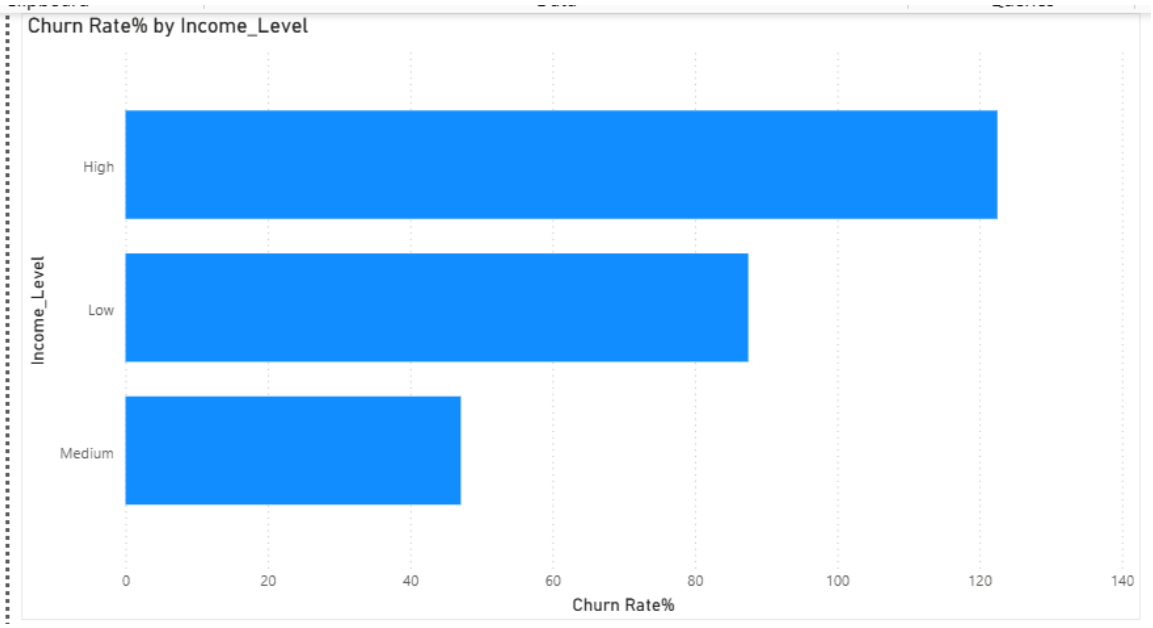
Screenshot 2.2: Churn Rate by Region



Analysis

Target should prioritize East-region retention strategies such as region-specific promotions, loyalty tier upgrades, and customer experience improvements, while sustaining and scaling successful practices from the Central and West regions.

Screenshot 2.3: Churn Rate by Income Level

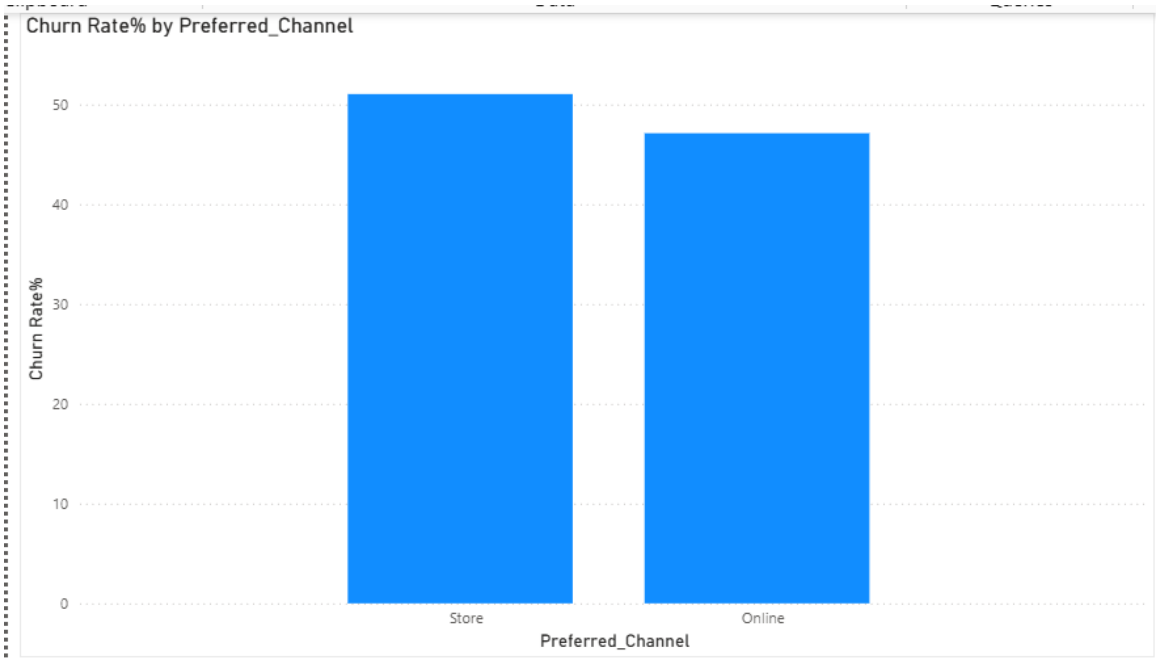


Analysis:

It shows that high-income customers have the highest churn rate, indicating that higher spending power does not necessarily translate into stronger loyalty. The low-income segment exhibits moderate churn, likely influenced by price sensitivity and reliance on

promotions. In contrast, medium-income customers have the lowest churn, suggesting better value alignment and satisfaction. This highlights an opportunity to improve retention among high-income users through personalized experiences rather than discounts. Overall, income-based segmentation can help tailor more effective retention strategies.

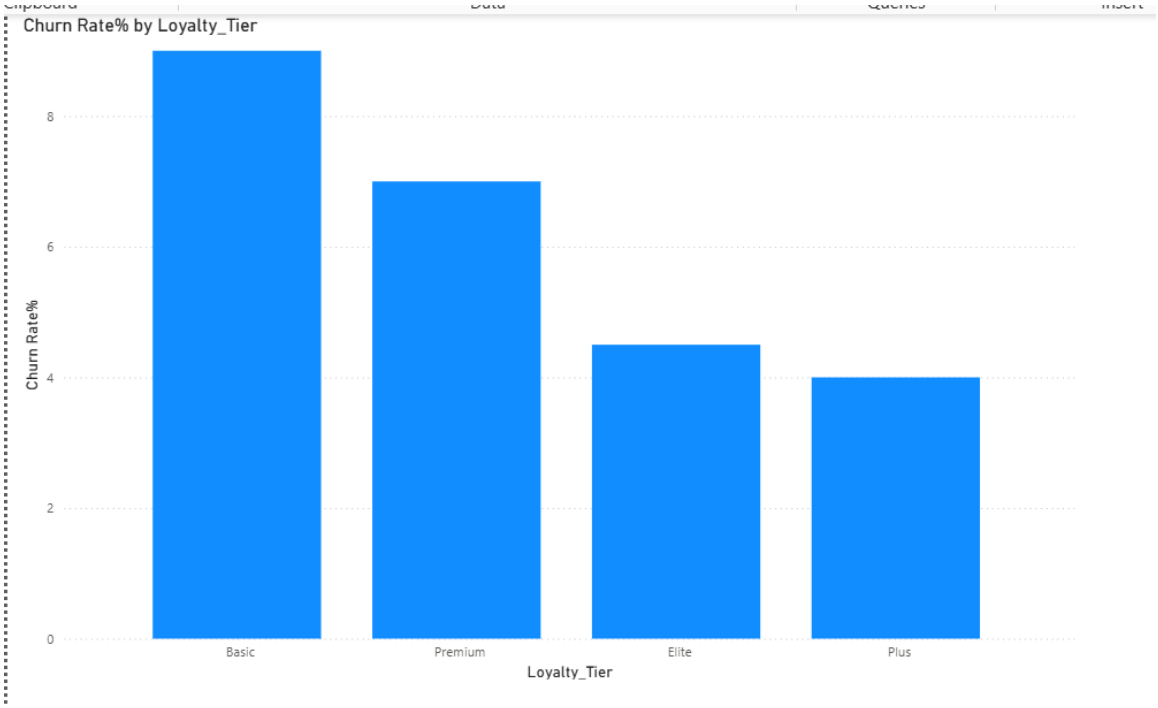
Screenshot 2.4: Churn Rate by Channel (Store vs Online)



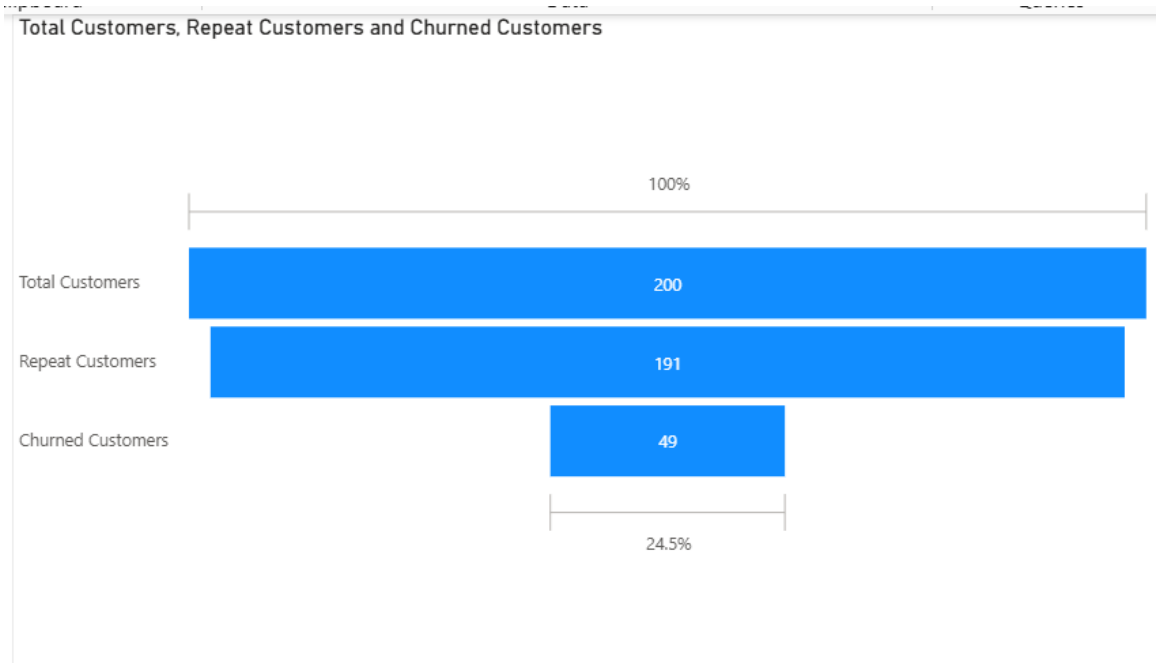
Analysis:

It shows that store-preferred customers have a slightly higher churn rate than online customers, indicating possible in-store experience or service consistency issues. Online customers demonstrate relatively better retention, likely due to convenience and personalized digital engagement. The small difference suggests that both channels face retention challenges, though the drivers may vary. Improving in-store loyalty programs and staff interactions could help reduce store churn. Aligning offline and online experiences will strengthen overall customer retention.

Screenshot 2.5: Churn Rate by Loyalty Tier



Analysis:
Churn rates show a clear inverse relationship with loyalty tiers, peaking at 9% for Basic members and dropping to 4% for Plus members. This data confirms that higher-tier status effectively drives retention. To improve overall stability, the business should prioritize reducing the high churn observed in the entry-level segment.
Screenshot 2.6: Customer Funnel (Total → Repeat → Churned)



Analysis:
The funnel visualization presents a total customer count of 200, representing the 100% baseline for the analyzed population. A high degree of loyalty is evident, with 191 repeat customers demonstrating consistent and recurring engagement with the brand. Conversely, the data identifies 49 churned customers, which translates to a significant 24.5% churn rate relative to the total base. This metric suggests that while the core repeat base is strong, nearly a quarter of the customer population has discontinued their relationship. Identifying the specific pain points for these 49 individuals is essential for stabilizing and growing the overall 200-person customer group.

Task 3: Repeat Purchase Analysis

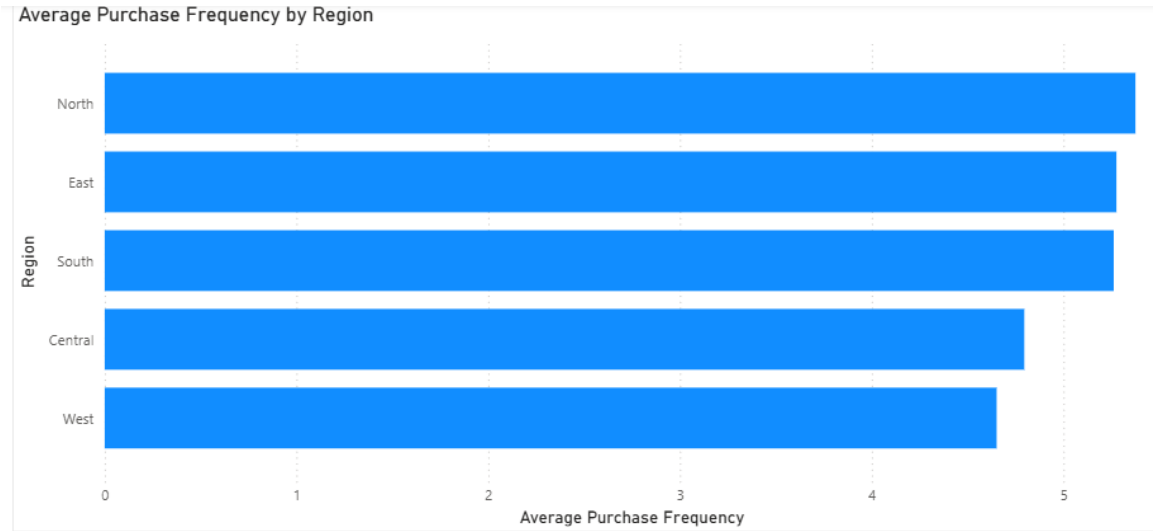
Customers were segmented into Low (0–3), Mid (4–8), and High (9+) purchase tiers. Average purchase frequency was compared across Region, Age Group, and Loyalty Tier. Top product categories purchased by loyal customers were identified.

Screenshot 3.1: Customer Purchase Frequency Segmentation

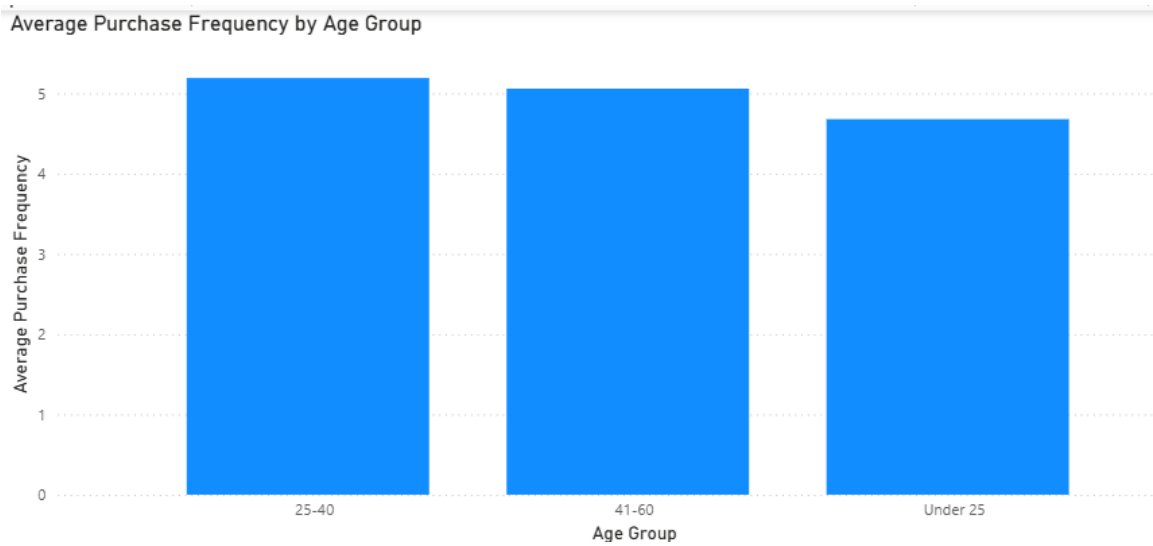
Purchase Segment	Cu
Mid-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
High Tier	Re
High Tier	Re
Mid-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
High Tier	Re
Mid-Tier	Re
High Tier	Re
Mid-Tier	Re
Low-Tier	Re
Mid-Tier	Re
Mid-Tier	Re
Low-Tier	Re
Mid-Tier	Re
High Tier	Re

```
Purchase Segment = SWITCH(TRUE(),Customer_Transactions[Purchase Count]<=3,"Low-Tier",Customer_Transactions[Purchase Count]<=8,"Mid-Tier","High Tier")
```

Screenshot 3.2: Average Purchase Frequency by Region



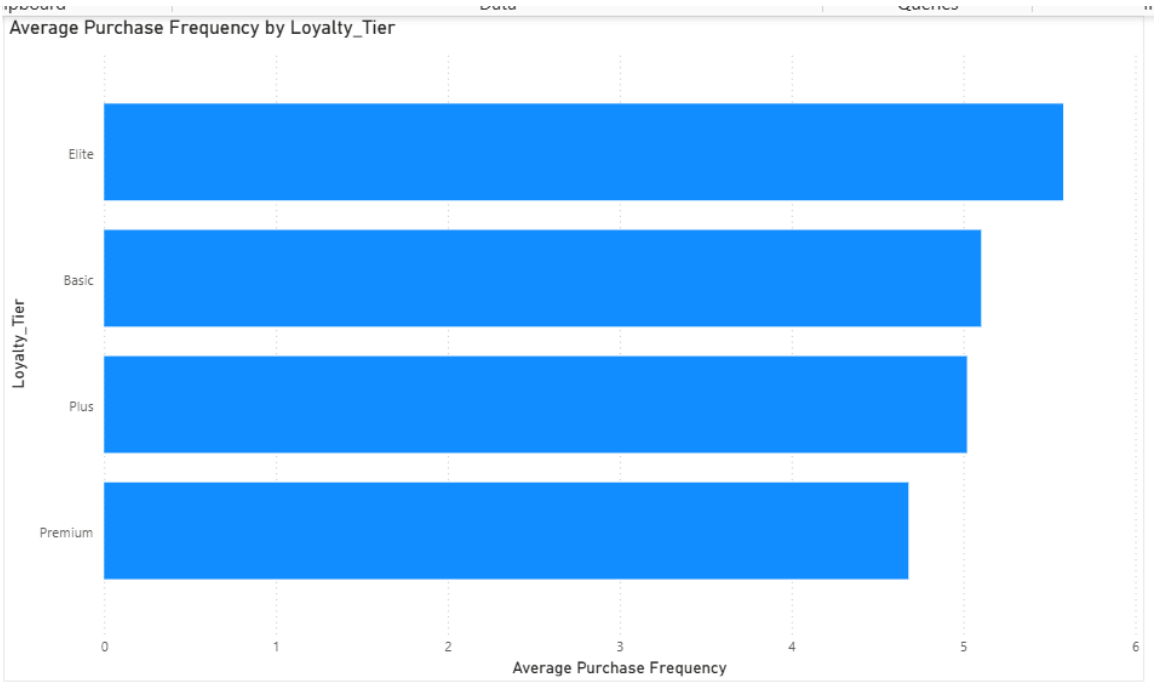
Analysis:
The organization manages **200 total customers**, retaining **191 repeat buyers** despite a **24.5% churn rate** affecting 49 individuals. Regional data shows the **North leads** in purchase frequency, followed closely by steady engagement in the East and South. In contrast, the Central and West regions lag behind, with the **West recording the lowest** average frequency. Addressing this churn while boosting engagement in underperforming regions like the West is essential for long-term stability. These combined insights suggest that targeted regional retention strategies could effectively maximize the value of the total customer base.
Screenshot 3.3: Average Purchase Frequency by Age Group



Analysis:
The business maintains a base of 200 total customers with 191 repeat buyers, yet faces a 24.5% churn rate involving 49 individuals. Regional performance shows that the North leads in average purchase frequency, while the West region currently trails all others.

Demographic data reveals that the 25-40 age group is the most active, whereas the "Under 25" segment has the lowest purchase frequency. These trends suggest that while core loyalty is high, significant engagement gaps exist across specific geographic and age-based segments. Prioritizing retention strategies for younger customers and those in the West could help stabilize the total customer population.

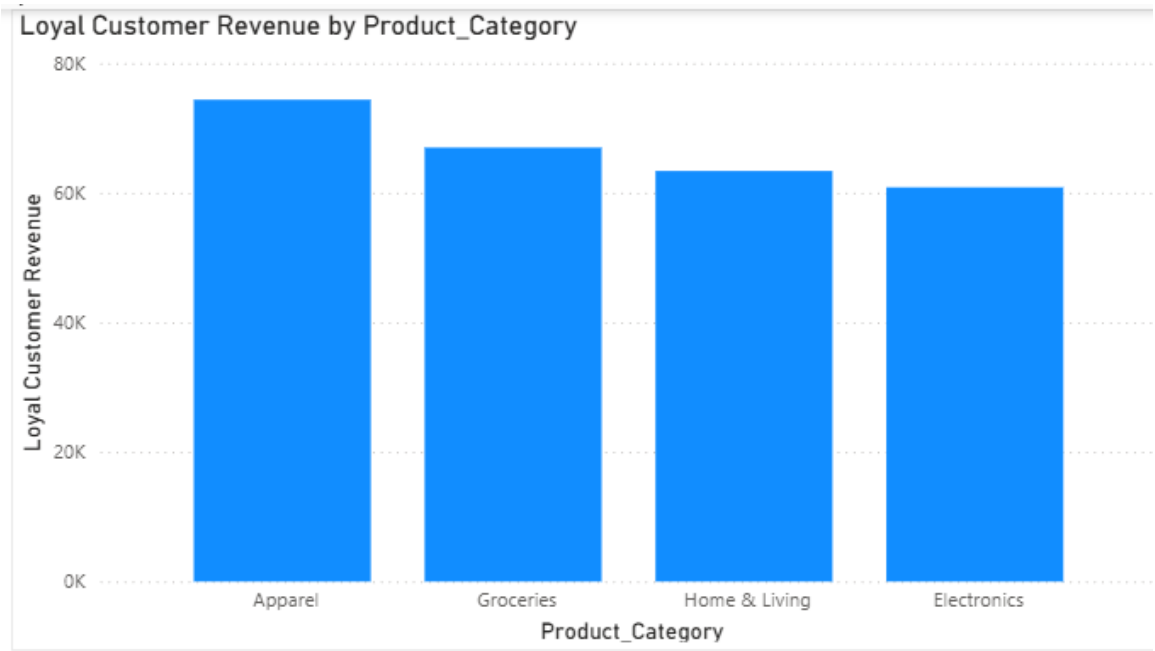
Screenshot 3.4: Average Purchase Frequency by Loyalty Tier



Analysis:

The chart shows that Elite customers have the highest average purchase frequency, indicating the strongest loyalty and repeat buying behavior. Basic and Plus tiers exhibit nearly identical purchase frequencies, suggesting similar engagement levels across these segments. Premium customers record the lowest average purchase frequency, highlighting comparatively weaker repeat behavior. Overall, purchase frequency tends to increase with higher loyalty tiers, validating the effectiveness of the loyalty structure. There is a clear opportunity to strengthen incentives for Premium and mid-tier customers to drive more frequent purchases.

Screenshot 3.5: Most Purchased Product Categories by Loyal Customers



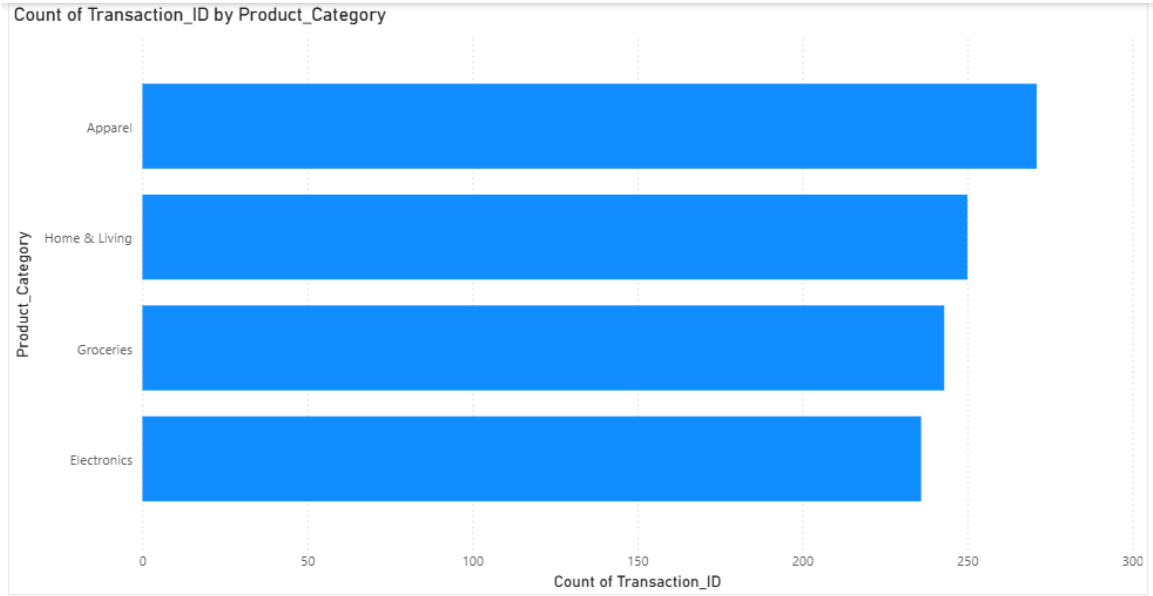
Analysis:

The chart indicates that Apparel generates the highest revenue from loyal customers, making it the strongest performing category. Groceries follow closely, showing consistent and repeat-driven spending behavior. Home & Living contributes moderately, reflecting steady but lower loyal customer revenue. Electronics has the lowest loyal customer revenue among the categories. Overall, the insights suggest that lifestyle and essential categories drive stronger loyalty-based revenue compared to high-involvement products like electronics.

Task 4: Promotion & Loyalty Impact

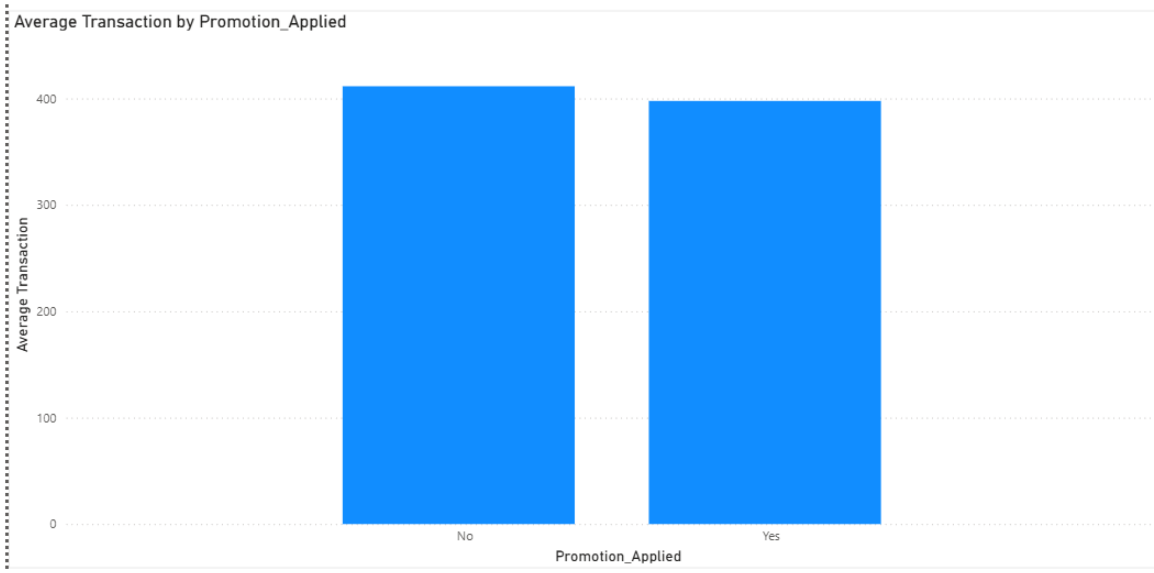
The percentage of transactions with promotions applied was calculated. Average transaction value with and without promotions was compared. Churn rate across loyalty tiers was analyzed. Points earned vs redeemed was visualized using a clustered column chart.

Screenshot 4.1: Percentage of Transactions with Promotion Applied



Analysis:
The chart shows that Apparel has the highest number of transactions, indicating strong customer demand in this category. Home & Living follows closely, suggesting it is another high-performing segment with consistent purchase activity. Groceries also records a substantial transaction count, reflecting frequent and essential buying behavior. Electronics has the lowest transactions among the categories but still maintains a solid volume. Overall, the distribution indicates relatively balanced performance, with Apparel leading slightly over other categories.

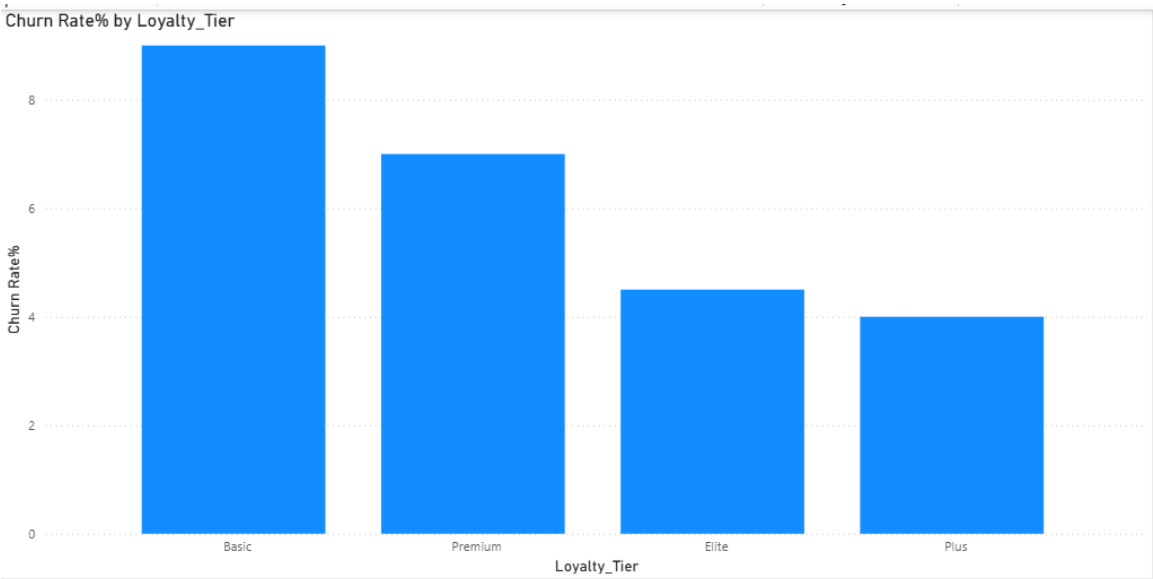
Screenshot 4.2: Average Purchase Amount (With vs Without Promotion)



Analysis:

The chart indicates that the average transaction value is slightly higher when no promotion is applied compared to when a promotion is used. Transactions without promotions show the highest average spend, suggesting customers may purchase higher-value items at full price. Promotional transactions, while marginally lower in average value, still remain close, indicating promotions do not significantly reduce order size. This implies promotions may be effective in driving volume rather than increasing basket value. Overall, the difference suggests promotions influence purchase behavior more on frequency than on spend per transaction.

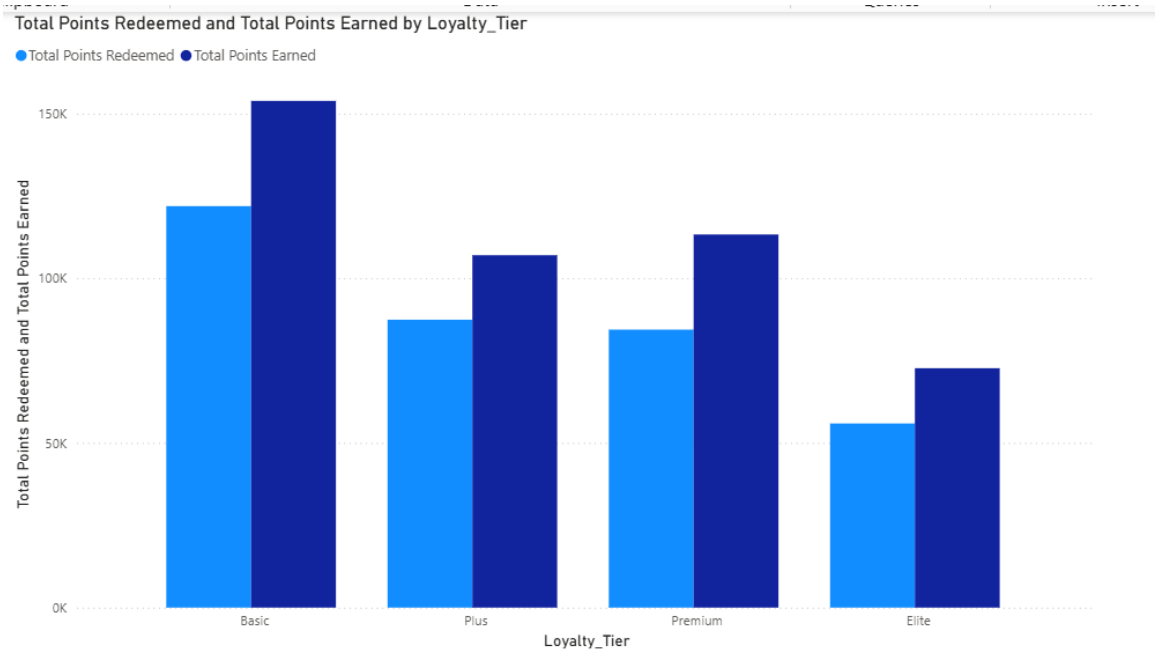
Screenshot 4.3: Churn Rate by Loyalty Tier



Analysis:

The chart shows that churn rate decreases as customers move up the loyalty tiers. Basic tier customers have the highest churn, indicating lower engagement and weaker retention. Premium customers exhibit moderate churn, suggesting improved but still vulnerable loyalty. Elite and Plus tiers have the lowest churn rates, reflecting strong customer satisfaction and retention. Overall, higher loyalty benefits are clearly associated with reduced customer churn.

Screenshot 4.4: Points Earned vs Points Redeemed by Loyalty Tier



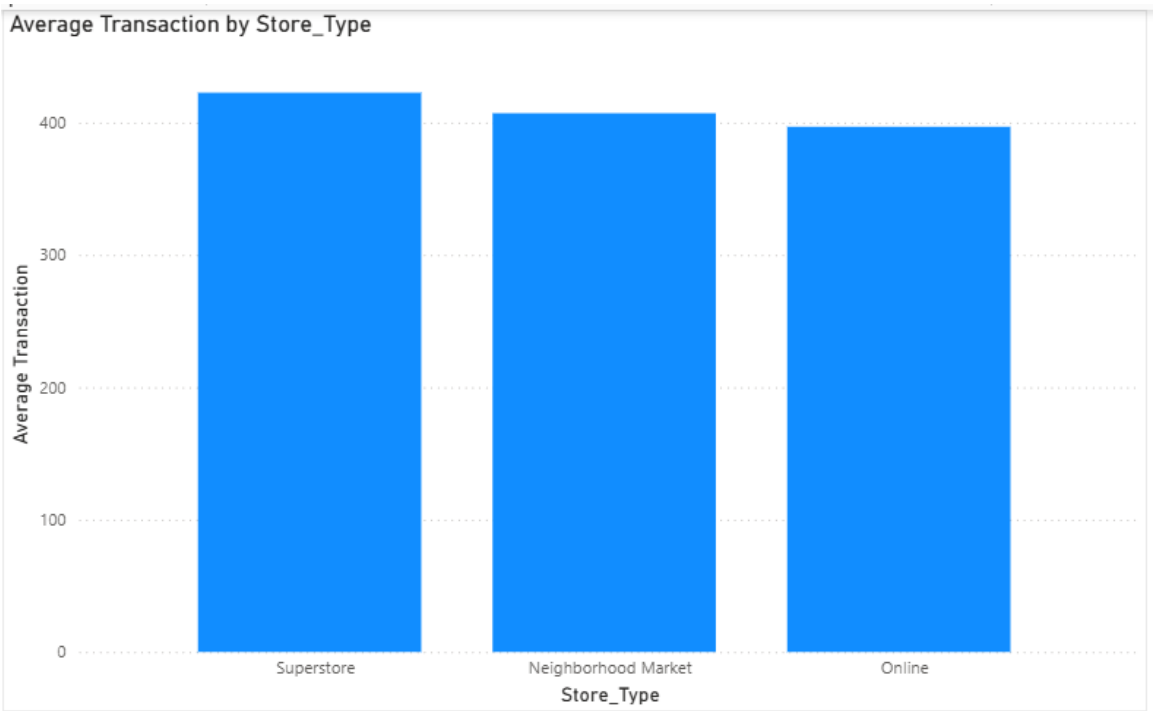
Analysis:

The chart shows that Basic tier customers earn and redeem the highest total loyalty points, driven by their large user base and frequent transactions. Premium and Plus tiers also demonstrate strong point activity, with earned points consistently higher than redeemed points, indicating ongoing engagement. Elite customers earn and redeem fewer total points, likely due to a smaller but more focused customer segment. Across all tiers, points earned exceed points redeemed, suggesting unused reward potential. Overall, higher tiers show quality engagement, while lower tiers drive volume in the loyalty program.

Task 5: Store & Channel Performance vs Retention

Store location data was merged with transaction data to analyze performance. Average transaction amount and churn rate were visualized by store type. The relationship between store opening year and customer retention was examined.

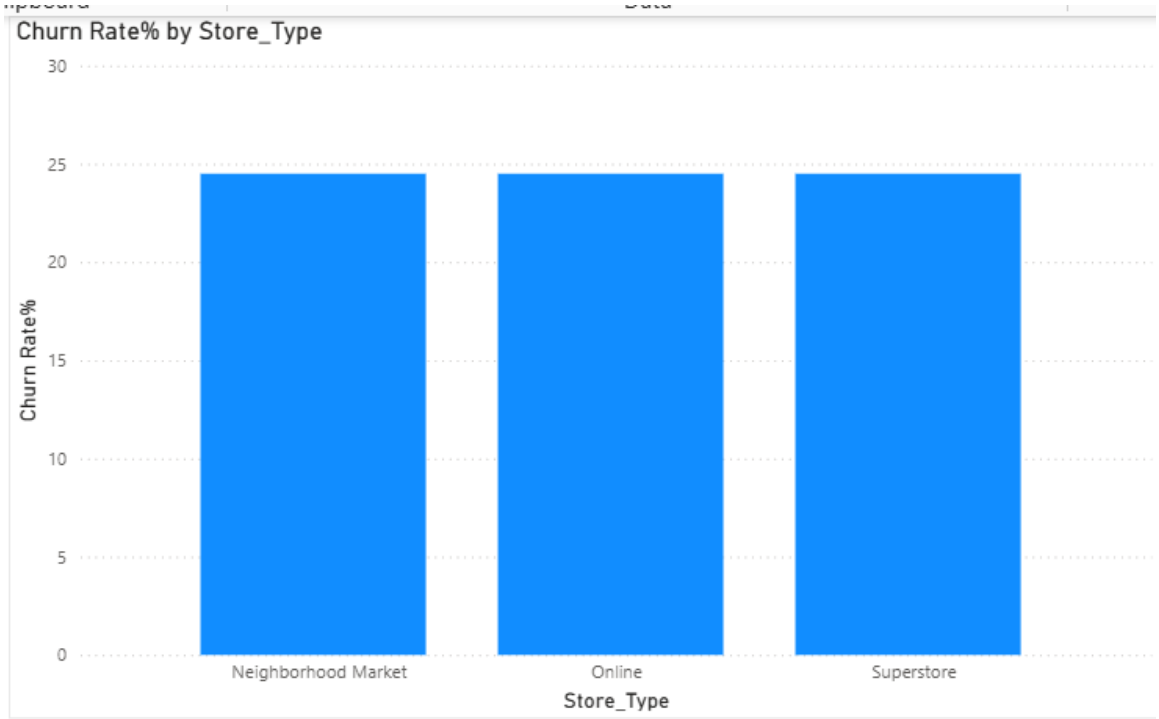
Screenshot 5.1: Average Transaction Amount by Store Type



Analysis:

The chart indicates that Superstores have the highest average transaction value, suggesting customers make larger basket purchases in these stores. Neighborhood Markets follow closely, reflecting steady spending driven by convenience-based shopping. Online transactions show a slightly lower average value, possibly due to smaller, more frequent purchases. The differences across store types are relatively moderate, indicating consistent spending behavior overall. This suggests an opportunity to increase online basket size through bundling or targeted offers.

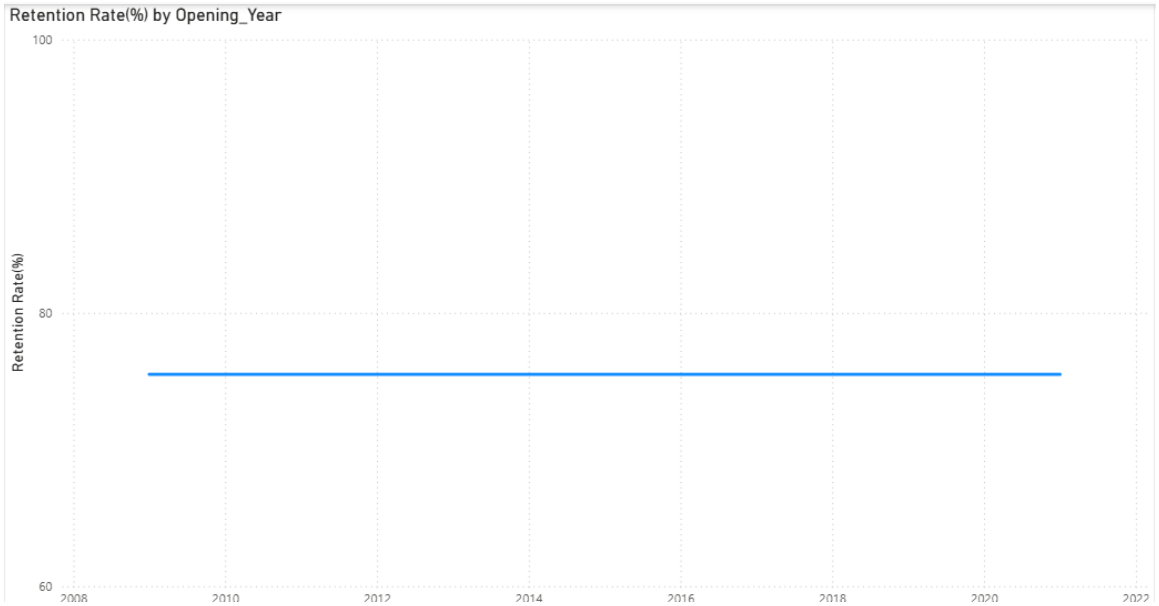
Screenshot 5.2: Churn Rate by Store Type



Analysis:

The chart shows that churn rates are almost identical across Neighborhood Market, Online, and Superstore channels. This indicates that customer retention is not strongly influenced by store type. All channels experience a similar level of customer drop-off, suggesting consistent satisfaction or pain points across formats. Since no channel stands out, retention strategies should be applied uniformly rather than channel-specific. Overall, churn appears to be driven more by customer-level factors than by the shopping channel.

Screenshot 5.3: Store Opening Year vs Customer Retention Analysis



Analysis:

The chart shows that the retention rate remains stable at around 75% across all opening years. This indicates that store age does not significantly impact customer retention. Both older and newer stores perform consistently in retaining customers. The flat trend suggests standardized operations and a uniform customer experience over time. Overall, retention appears driven by broader business factors rather than the store's opening year.

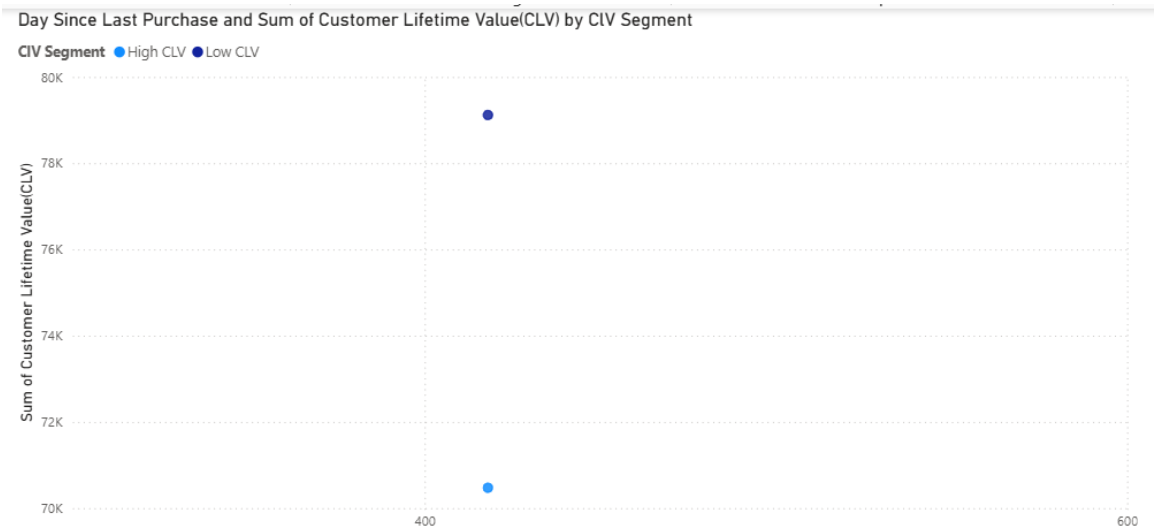
Task 6: Customer Lifetime Value (CLV) Analysis

Customer Lifetime Value (CLV) was calculated as total spend divided by membership duration in years. Customers were segmented into High and Low CLV groups based on average CLV. Visuals compared CLV with recency and across loyalty tiers and regions.

Screenshot 6.1: CLV Calculation Table

```
Customer Lifetime Value(CLV) = DIVIDE([Total Amount Spent],Customer_Demographics[Membership Duration(Years)])
```

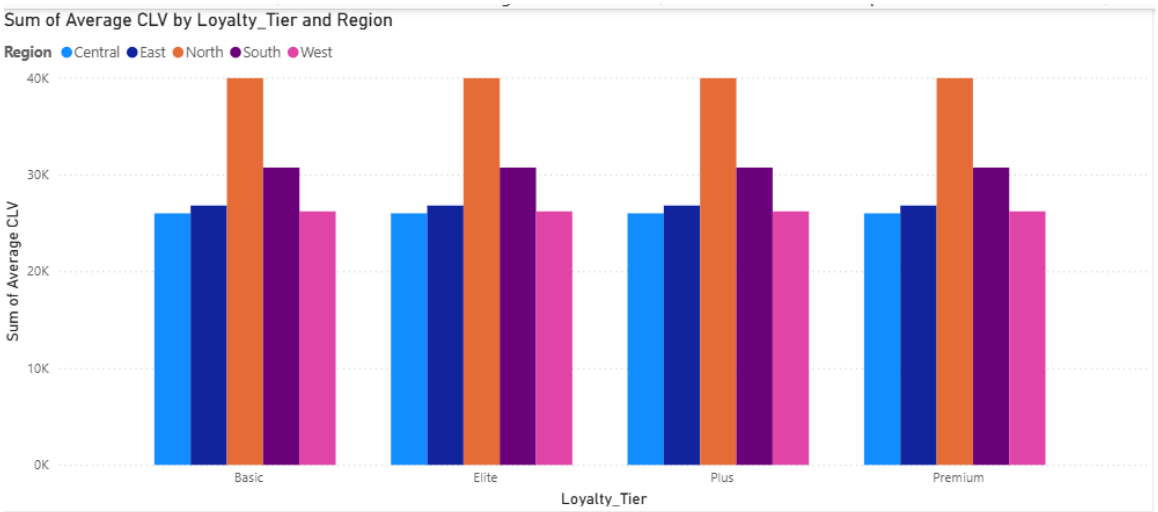
Screenshot 6.2: CLV vs Days Since Last Purchase



Analysis:

The chart shows a clear separation between High CLV and Low CLV customer segments in terms of total lifetime value. High CLV customers contribute significantly more to overall revenue despite having similar days since last purchase. Low CLV customers show a noticeably lower cumulative value, indicating limited long-term contribution. The similar recency suggests value differences are driven more by spending behavior than inactivity. This highlights the importance of prioritizing engagement and retention strategies for high CLV customers.

Screenshot 6.3: CLV by Loyalty Tier And Region



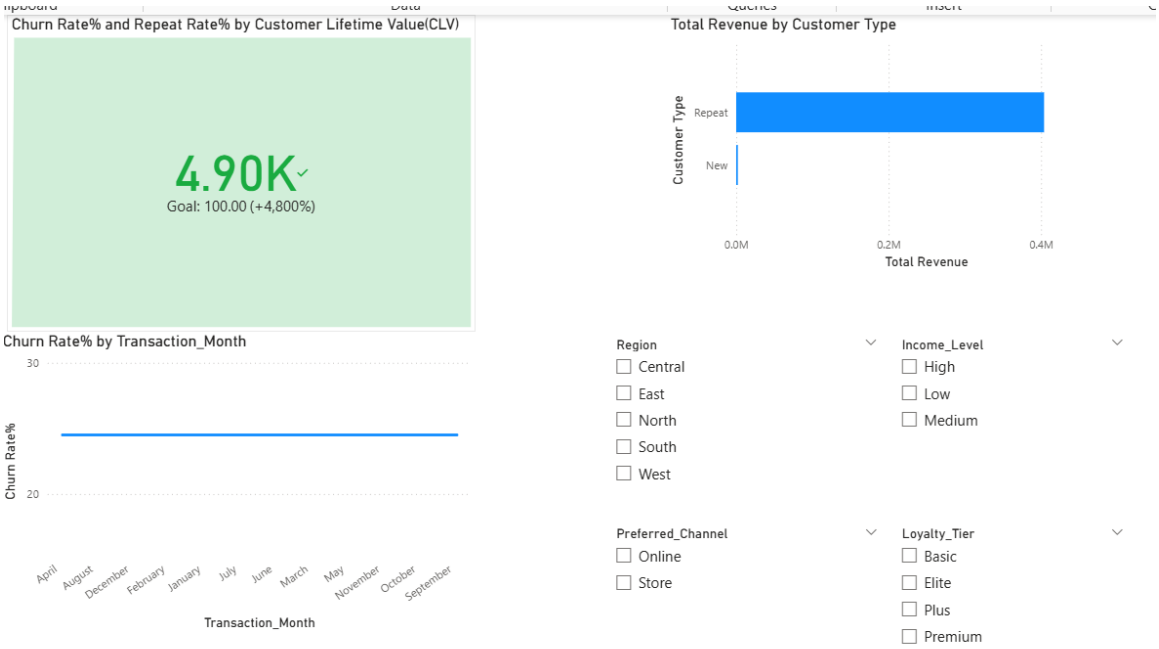
Analysis:

The chart shows that the North region consistently has the highest average CLV across all loyalty tiers, indicating stronger customer value compared to other regions. South follows as the second-highest contributor, while Central and West remain relatively lower and stable across tiers. Average CLV values are fairly uniform within each region, suggesting loyalty tier upgrades do not significantly change CLV by themselves. This implies regional factors have a stronger impact on CLV than loyalty segmentation. Overall, focusing retention and upsell strategies in high-performing regions like North and South could yield better returns.

Task 7: Final Dashboard & Executive Summary

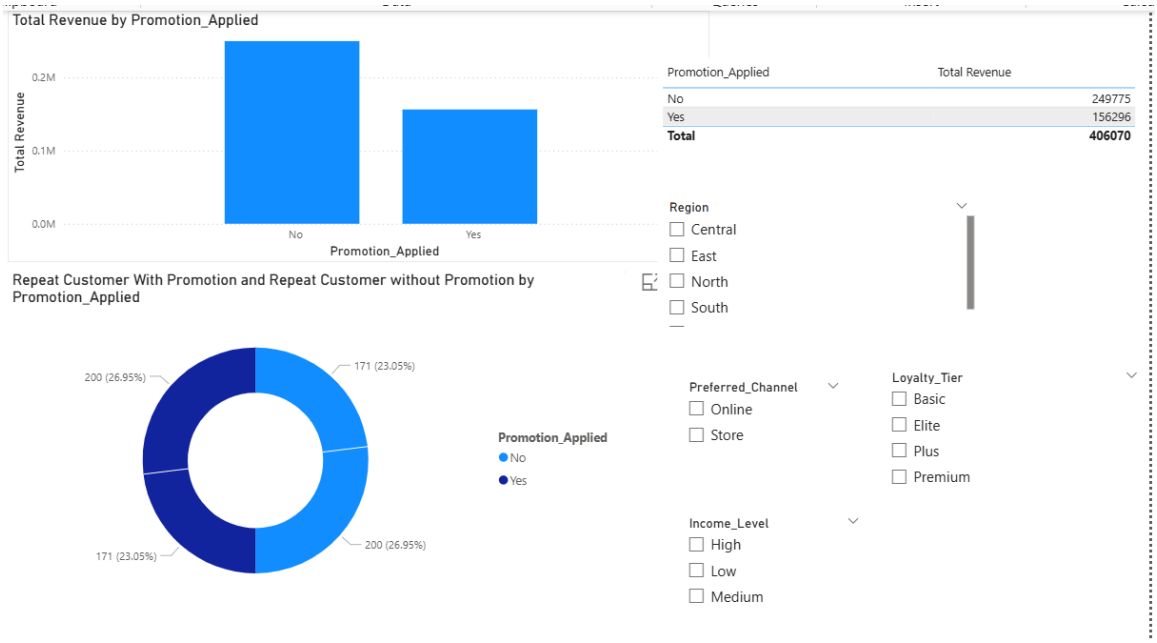
A multi-page Power BI dashboard was developed covering KPIs, loyalty and promotion impact, store/channel insights, and customer segmentation. Interactive slicers were added for Region, Channel, Income Level, and Loyalty Tier. Key business recommendations were derived from insights.

Screenshot 7.1: Dashboard Page 1 – Key KPIs (Churn, CLV, Repeat Rate)



Analysis:

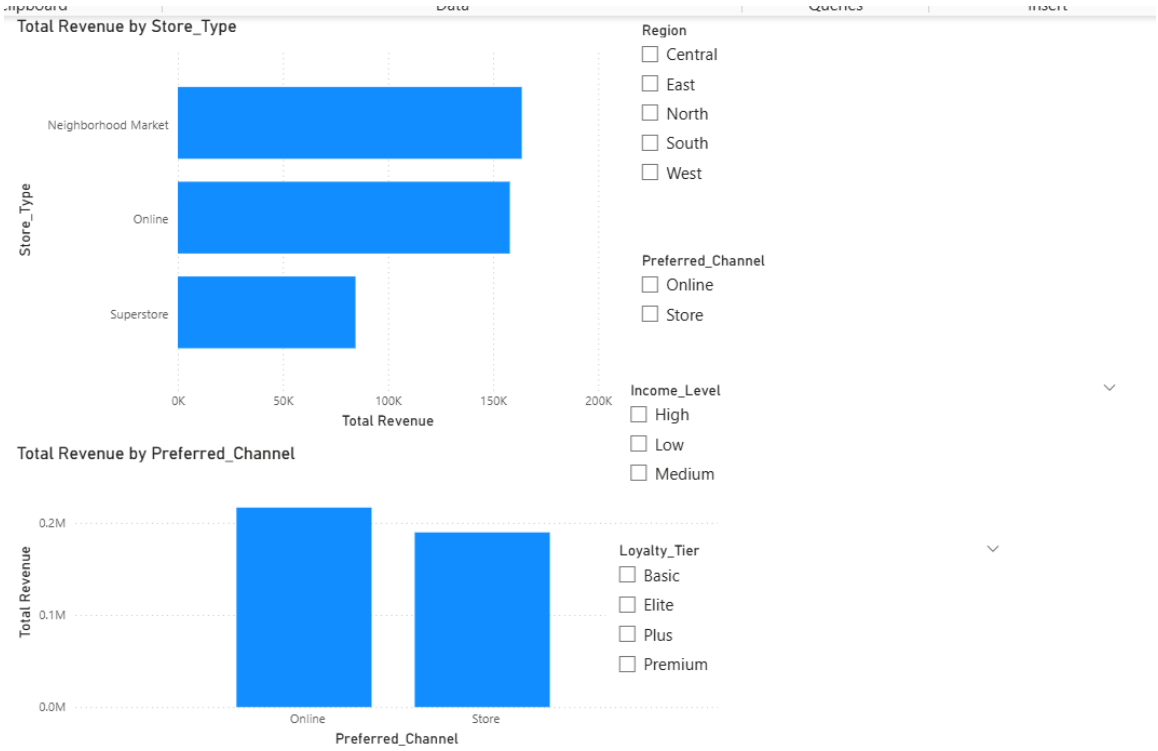
The dashboard indicates a very high combined repeat and churn metric (4.90K), far exceeding the target, which suggests strong customer activity driven largely by repeat customers. Total revenue is overwhelmingly contributed by repeat customers, while new customers add only a minimal share. The churn rate remains almost flat across all months, indicating stable but unimproving retention performance. This stability suggests existing retention strategies are consistent but not reducing churn over time. Overall, revenue growth appears more dependent on repeat customers than on new customer acquisition.
Screenshot 7.2: Dashboard Page 2 – Loyalty & Promotion Impact



Analysis:

The dashboard shows that higher total revenue is generated when promotions are not applied compared to when promotions are used. Despite this, repeat customers are fairly evenly split between promotion and non-promotion scenarios, indicating promotions do not drastically change repeat behavior. This suggests that loyal customers may purchase regardless of discounts. Promotions appear to contribute incremental revenue rather than driving the majority of sales. Overall, optimizing promotions for targeted segments rather than broad application could improve revenue efficiency.

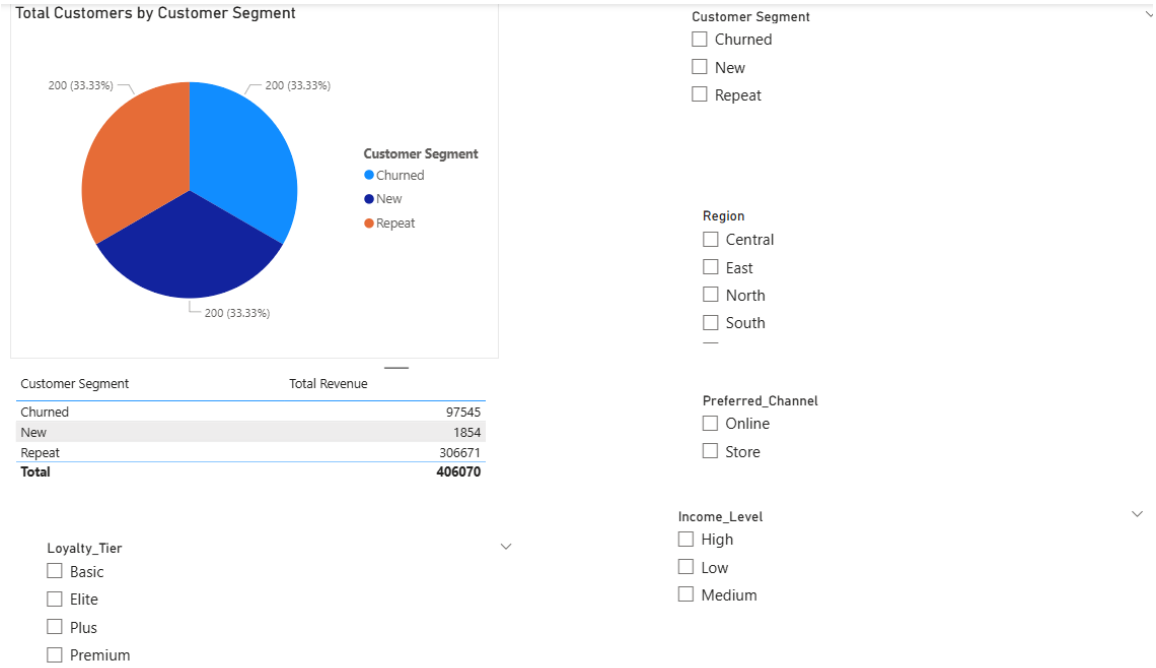
Screenshot 7.3: Dashboard Page 3 – Store & Channel Insights



Analysis:

The analysis shows that Neighborhood Markets generate the highest total revenue, closely followed by the Online store type, while Superstores contribute the least. When viewed by channel, online purchases outperform in-store sales, indicating a strong customer preference for digital shopping. This suggests convenience and accessibility are key revenue drivers. Physical stores still contribute significantly but lag behind online performance. Strengthening the online experience and selectively optimizing underperforming store formats could further improve overall revenue.

Screenshot 7.4: Dashboard Page 4 – Customer Segmentation View



Answer:

The customer base is evenly split across churned, new, and repeat segments, each contributing roughly one-third of total customers. However, repeat customers generate the highest revenue, highlighting their stronger lifetime value. Churned customers still account for a significant revenue share, suggesting recent churn or high past spend. New customers contribute minimal revenue, indicating early-stage or low initial purchases. Overall, improving new customer activation and preventing churn could significantly boost total revenue.

Conclusion

This project enables Target to identify churn drivers, high-value customers, and underperforming channels. The insights support data-driven strategies to improve customer retention, loyalty engagement, and long-term profitability.